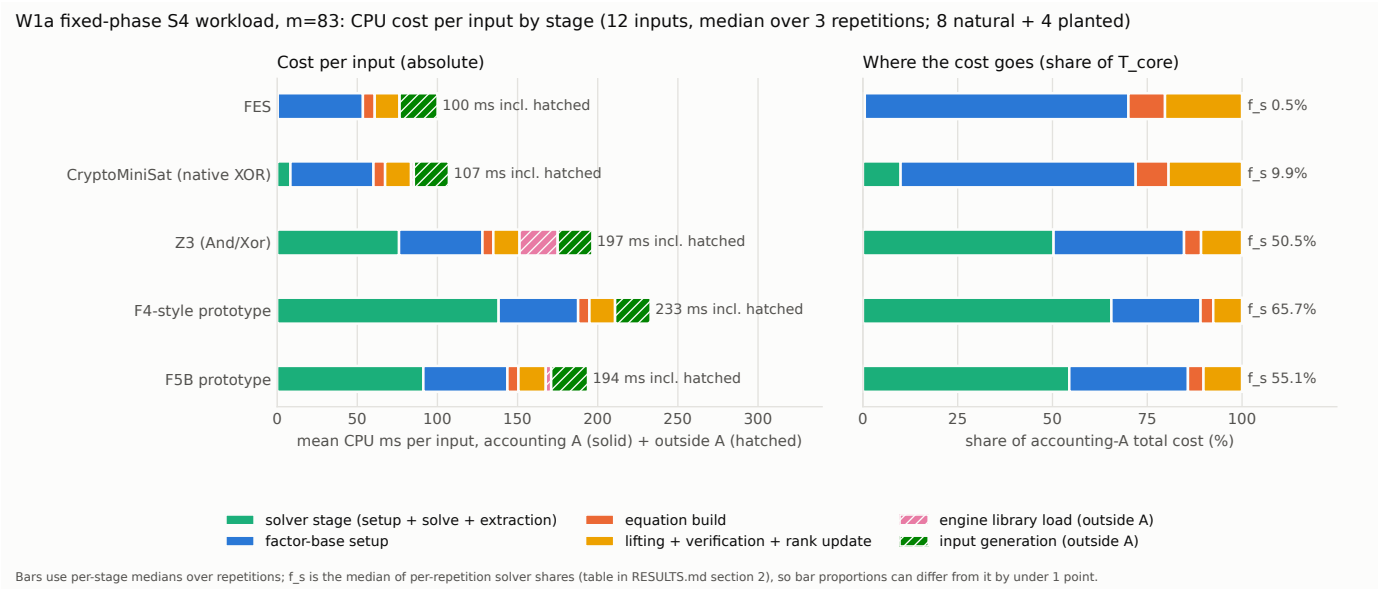
Total verified-relation cost of FES, SAT, F4-style and F5B solvers (Lead 4)

Directory `research/notes/ecc2k130/solver_total_cost_20261011/`. Protocol: `PROTOCOL.md`, committed (`fdb7692f4`) before any frozen measurement. Base commit `bbd330a9d` (origin/main as of 2026-10-11; it could not be refreshed afterwards). Every number in the tables and sentences below is rendered from `results/summary.json` by `summarize.py`; `verify.py` recomputes the summary from the raw receipts and checks that this file is exactly that rendering.

This is a stage-composition diagnostic. No discrete logarithm was recovered. A matched Pollard-rho reference, `S = total_operations / sqrt(r)`, the ratio to the floor, any end-to-end speedup and any ECC2K-130 transfer are **not measured and are left null**. Under `AGENTS.md` section 3 the work is classified **accounting**: the algorithms did not change, only what is charged to them.



Bars use per-stage medians over repetitions; f_s is the median of per-repetition solver shares (table in RESULTS.md section 2), so bar proportions can differ from it by under 1 point.

0. Requested deliverables and status

requirement	status	evidence
Read the m83 PROTOCOL/RESULTS and runners; preflight passes; record engines	verified complete	`results/preflight.json`: frozen `verify_archive.py` and `preflight_engines.py` pass on a scratch copy; `pycryptosat` 5.16.0 and Z3 5.1.0 available; msolve, WDSat, Sage and the `cryptominisat5` CLI unavailable here
Protocol written and committed before measuring	verified complete	`PROTOCOL.md`, commit `fdb7692f4`
W1: every engine, every input, CPU and wall seconds per stage, planted rows charged but not counted	verified complete	`results/w1/receipts.jsonl` (360 children = 3 repetitions x 120 cells)
Primary metric: seconds per newly independent verified relation	**undefined for the measured workloads (0 natural relations)** reported as cost per attempt, cost per verified planted relation, a measured 95% bound and a labelled derived bound (section 5)	table "cost per relation"
Solver-stage fraction of total	verified complete, with its sensitivity to accounting (sections 2-4)	tables, `figures/`
RESULTS.md, stacked-bar figure, PDF	complete; figures are SVG (+PNG); PDF rendered with Typst (`results/report_build.json`)	`figures/`, `report.pdf`
Verify script that recomputes the summary	verified complete	`verify.py`, `results/verify_report.json`
W2: n=13 CryptoMiniSat vs Z3, 120 s deadlines	verified complete: 8 children (4 targets x 2 engines); the two Z3 deadline timeouts are preserved and charged	`results/w2/receipts.jsonl`
Pull request	**not opened**: GitHub access was unavailable (raw git 403, API 404). Committed locally; body prepared	local branch

1. What ran

- **W1a**: the 12 frozen fixed-phase S4 inputs of the m83 matrix (seeds 260938 and 260939, 2 planted plus 4 natural per seed), 5 engines, 3 repetitions: 180 children.
- **W1b**: the normal-subspace payload sweep s = 2, 3, 4 (planted input 0 and natural input 2 per seed), 5 engines, 3 repetitions: 180 children.
- Per-child limits as in the m83 protocol: 1 GiB address space, 70 s wall kill, 25 s equation-build deadline, 5 s aggregate solver deadline, one benchmark process at a time. The host (4 cores, shared with other workers) had 1-minute load averages up to 2.317 at a W1a child start.
- Every completed cell reproduces the input hash, equation hash, complete algebraic root set and group verification of the same-repetition FES receipt (255 cells) and of the archived m83 receipt (246 cells); there are no mismatches (`results/verify_report.json`).

2. Solver-stage ranking versus total-cost ranking (W1a)

Engine (W1a, 12 inputs)	complete	natural relations / natural attempts	planted verified	solver stage ms/input (CPU)	total T_core ms/input (CPU)	solver share f_s	Amdahl bound g
FES	12/12	0/8	4/4	0.38	79.21	0.5%	1.00x
CryptoMiniSat (native XOR)	12/12	0/8	4/4	8.31	82.15	9.9%	1.11x
Z3 (And/Xor)	12/12	0/8	4/4	76.04	154.85	50.5%	2.02x
F4-style prototype	12/12	0/8	4/4	138.18	211.40	65.7%	2.91x
F5B prototype	12/12	0/8	4/4	91.22	167.48	55.1%	2.22x

`f_s` is the solver stage (solver setup + solve + model extraction) as a fraction of `T_core`, the sum of all charged stages (accounting A: factor-base setup, equation build, solver stage, lifting, verification, rank update). `g = T_core / (T_core - T_solver)` is the most that making the solve stage free could save. Natural relations are new independent verified natural relations over all natural attempts; planted inputs are lifted, verified and rank-updated (their cost is charged) but never counted.

W1a ranking comparison	value
order by solver stage (T_solver, CPU)	FES < CryptoMiniSat (native XOR) < Z3 (And/Xor) < F5B prototype < F4-style prototype
order by total cost (T_core, A, CPU)	FES < CryptoMiniSat (native XOR) < Z3 (And/Xor) < F5B prototype < F4-style prototype
Kendall tau between the two orderings	1.00
order by T_core under B (setup amortised)	FES < CryptoMiniSat (native XOR) < Z3 (And/Xor) < F5B prototype < F4-style prototype
order by T_core under C (fully loaded)	FES < CryptoMiniSat (native XOR) < F5B prototype < Z3 (And/Xor) < F4-style prototype
Kendall tau, solver stage vs T_core under B / under C	1.00 / 0.80
max/min over engines, solver stage	365.4x
max/min over engines, total T_core (A)	2.7x
max/min over engines, m83-definition solver stage (incl. library load)	567.7x
post hoc steady state: max/min over engines, T_ss	19.9x
post hoc steady state: Kendall tau, solver stage vs T_ss	1.00

Under accounting A the two orderings are identical (Kendall tau 1.000): the non-solver stages are common to all engines, and an additive common cost cannot reorder them. What changes is the size of the gaps. The solver stages span 365.4x (max/min over engines); total cost spans 2.7x. CryptoMiniSat's solver stage is 22.0x FES's, but its total cost is 1.04x FES's (accounting A, CPU seconds). Z3's total is 1.88x CryptoMiniSat's, against 9.1x in the solver stage alone.

3. Which component dominates (W1a)

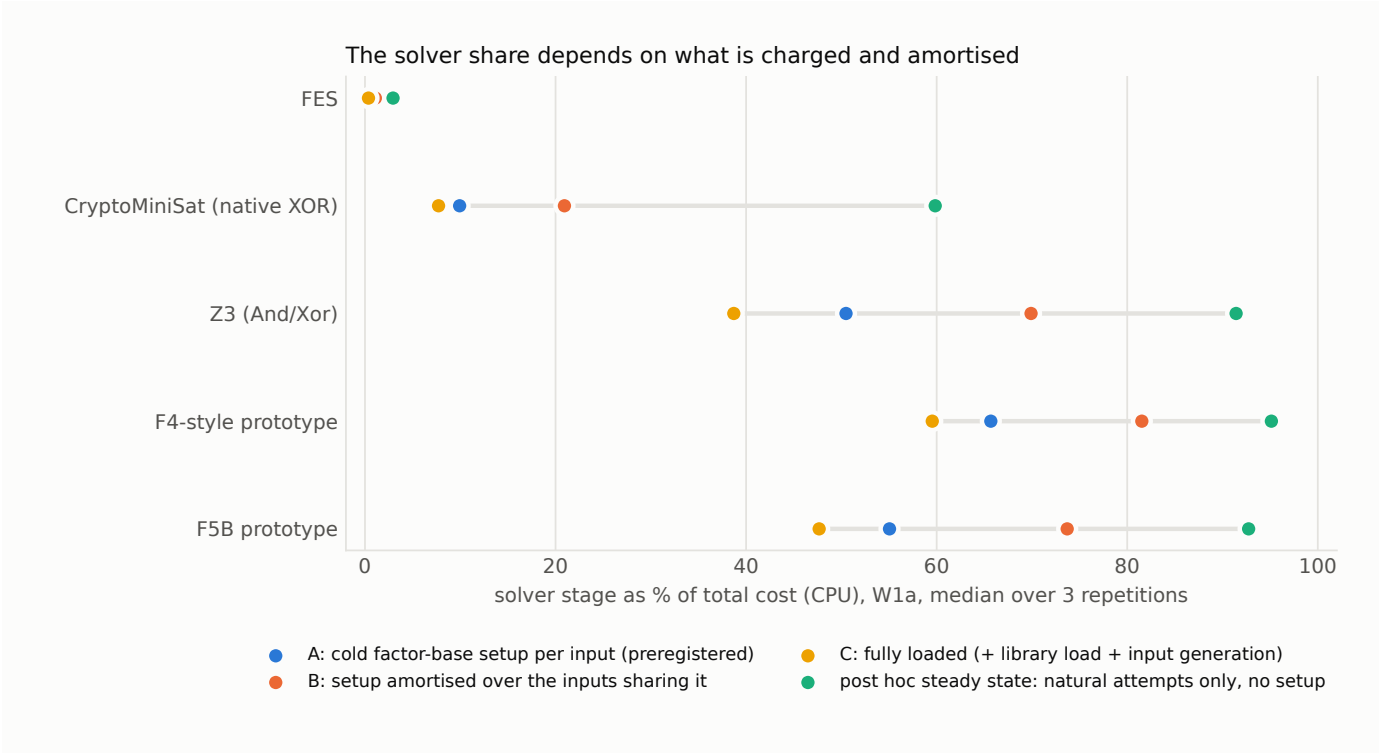
Engine (mean CPU ms per input)	fb setup	input gen	eq build	engine load	solver setup	solve	model extr.	lifting	verif.	rank upd.	T_core (A)
FES	53.140	23.686	7.371	0.000	0.000	0.304	0.074	10.443	5.076	0.011	79.21
CryptoMiniSat (native XOR)	51.804	21.874	7.249	1.728	0.677	7.468	0.160	10.336	5.843	0.012	82.15
Z3 (And/Xor)	52.039	21.638	6.793	23.795	66.282	9.343	0.430	10.900	5.440	0.012	154.85
F4-style prototype	49.585	22.153	7.050	0.000	0.000	137.528	0.661	10.634	5.378	0.013	211.40
F5B prototype	52.447	22.985	6.875	3.369	0.283	90.815	0.128	11.057	5.977	0.013	167.48

- FES and CryptoMiniSat: factor-base setup is the largest charged stage (53.1 ms of 79.2 ms for FES; 51.8 ms of 82.2 ms for CryptoMiniSat). Their solver stages are 0.38 ms and 8.31 ms.
- Lifting, verification and rank update are spent only on the 4 planted inputs (their algebraic roots, including the 2 that failed to lift). All 8 natural targets returned an **empty root set** from every engine: their Boolean systems are unsatisfiable, so a natural attempt incurs no lifting cost at all. Averaged over all 12 inputs, lifting + verification cost 10.4 ms + 5.1 ms per input, more than the whole FES solve.
- Z3: the largest stage is solver setup (translating the equations to And/Xor terms, 66.3 ms), then the solve (9.3 ms) and library load (23.8 ms, outside accounting A).
- F4-style and F5B prototypes: the solve stage dominates (137.5 ms and 90.8 ms).
- Equation build is 7.4 ms for every engine (same equations); rank updates cost about 0.011 ms.

4. How much of this is the accounting

Engine (W1a)	A: core, cold ms/input	f_s (A)	B: setup amortised ms/input	f_s (B)	C: fully loaded ms/input	f_s (C)	A in wall ms/input	f_s (A, wall)
FES	79.21	0.5%	32.85	1.1%	102.90	0.4%	80.16	0.5%
CryptoMiniSat (native XOR)	82.15	9.9%	39.89	20.9%	105.70	7.7%	83.11	10.0%
Z3 (And/Xor)	154.85	50.5%	108.77	69.9%	200.30	38.7%	159.04	50.4%

F4-style prototype	211.40	65.7%	170.08	81.5%	233.55	59.5%	213.92	65.5%
F5B prototype	167.48	55.1%	123.77	73.7%	193.83	47.7%	168.89	55.6%



Accounting A (preregistered primary) charges every child its own cold factor-base setup. B divides that setup over the inputs sharing a factor base (6 per seed for W1a). C adds library load and input generation. A fourth view was **added after seeing the first results and is not preregistered**: a steady-state stream of natural attempts with no factor-base setup, library load or input generation:

Engine (W1a, 8 natural attempts; post hoc, not preregistered)	T_ss CPU ms per attempt	of which solver stage ms	solver share f_s	Amdahl bound g
FES	7.31	0.22	2.9%	1.03x
CryptoMiniSat (native XOR)	17.72	10.26	59.8%	2.49x
Z3 (And/Xor)	81.86	74.26	91.4%	11.67x
F4-style prototype	145.83	138.73	95.1%	20.53x
F5B prototype	96.98	89.94	92.7%	13.78x

The engine ordering is the same under A, B and the post hoc steady state (Kendall tau 1.000 under B and 1.000 post hoc). Under C, which also charges library load, Z3 and F5B exchange places (tau 0.800): Z3's 23.8 ms import is charged there. The size of the solver share is the accounting-dependent quantity: FES stays at or below 2.9% while CryptoMiniSat moves from 9.9% (A) to 20.9% (B) to 59.8% (post hoc steady state). In that steady state the spread over engines is 19.9x instead of 2.7x, and CryptoMiniSat costs 2.42x FES per natural attempt instead of 1.04x. Whether the solver choice among FES and CryptoMiniSat matters therefore depends on how many inputs share one factor-base setup, a pipeline property this workload does not fix.

5. Yield: zero natural relations, so cost per relation is undefined here

Engine (W1a)	new independent natural relations / attempts	CPU ms per natural attempt (T_core)	CPU ms per verified planted relation	95% upper bound on yield/attempt	CPU s per relation, measured 95% lower bound	CPU s per relation, derived lower bound (uniform targets)
FES	0/8	58.53	107.09	0.312	0.187	1.01e+21
CryptoMiniSat (native XOR)	0/8	68.65	109.15	0.312	0.220	1.19e+21
Z3 (And/Xor)	0/8	138.29	185.20	0.312	0.443	2.39e+21
F4-style prototype	0/8	195.13	243.93	0.312	0.625	3.37e+21
F5B prototype	0/8	151.30	195.29	0.312	0.484	2.61e+21

- Measured: no engine produced a natural relation in the 8 natural attempts (the m83 full-orbit control over all 83 phases found none either). The primary metric, seconds per newly independent verified relation, is therefore **undefined (infinite) at this factor-base size, and that is itself the finding**. Natural rank is 0 for every engine.
- Reported instead: T_core per natural attempt; T_core per verified planted relation (4 of 4 planted controls verified for every engine; planted cost includes lifting, verification and a charged rank update on a scratch matrix); and a measured one-sided 95% Clopper-Pearson bound: 0 successes

in 8 attempts bounds the yield per attempt below 0.312, so cost per relation is at least the per-attempt cost divided by 0.312. That bound is tiny next to the derived value below, because 8 attempts cannot distinguish a yield of 0.3 from 1e-22.
- **Derived, not measured**: for a uniformly random target in the 81-bit subgroup the chance that a fixed-phase attempt hits one of the ordered triples of signed factor-base lifts is at most `lifts^3 / (r - 1)` (64 and 216 triples for the two seeds, `r = 241785163923079621668568 9`). Dividing the measured per-attempt cost by that probability gives the last column, of order 1e21 CPU seconds per relation. Every engine sees the same equations and the same roots, hence the same yield, so the total-cost ranking per relation equals the per-attempt ranking of section 2; no engine choice rescues a workload whose expected yield is about 6e-23 per attempt.

6. Payload sweep, s = 2, 3, 4 (W1b)

s	Engine	complete / timeout / setup_failed (cells, median over reps)	cells whose status changed between repetitions	solver ms/attempt (CPU)	T_core ms/attempt (CPU)	f_s	natural relations / completed natural
s2	FES	2/0/2	-	0.21	35.17	0.6%	0/2
s2	CryptoMiniSat (native XOR)	2/0/2	-	10.16	45.26	24.1%	0/2
s2	Z3 (And/Xor)	2/0/2	-	69.49	106.01	68.0%	0/2
s2	F4-style prototype	2/0/2	-	131.68	172.57	79.3%	0/2
s2	F5B prototype	2/0/2	-	80.89	124.00	65.6%	0/2
s3	FES	2/0/2	-	6.53	78.24	8.3%	0/2
s3	CryptoMiniSat (native XOR)	2/0/2	-	12.15	84.06	14.7%	0/2
s3	Z3 (And/Xor)	2/0/2	-	443.41	532.06	84.3%	0/2
s3	F4-style prototype	0/2/2	-	4934.21	5002.51	98.6%	0/0
s3	F5B prototype	0/2/2	s3-260939-i2	4917.70	4993.55	98.5%	0/0
s4	FES	3/0/1	-	234.93	507.80	46.3%	0/2
s4	CryptoMiniSat (native XOR)	3/0/1	-	152.95	436.02	35.1%	0/2
s4	Z3 (And/Xor)	3/0/1	s4-260939-i0	4493.91	4747.15	94.2%	0/2
s4	F4-style prototype	0/3/1	-	4936.27	5175.44	95.5%	0/0
s4	F5B prototype	0/3/1	-	5252.37	5470.51	96.0%	0/0

s	Engine (mean CPU ms per attempt)	fb setup	input gen	eq build	engine load	solver setup	solve	model extr.	lifting	verif.	rank upd.	T_core (A)
s3	FES	37.14	22.26	34.57	0.00	0.00	6.52	0.01	0.00	0.00	0.00	78.2
s3	CryptoMiniSat (native XOR)	37.27	21.54	33.43	1.72	2.60	9.56	0.01	0.00	0.00	0.00	84.1
s3	Z3 (And/Xor)	44.93	28.15	34.49	27.80	311.75	130.26	0.01	0.00	0.00	0.00	532.1
s3	F4-style prototype	37.14	20.35	32.68	0.00	0.00	4934.21	0.00	0.00	0.00	0.00	5002.5
s3	F5B prototype	37.37	21.70	34.27	7.04	1.27	4916.14	0.00	0.00	0.00	0.00	4993.6
s4	FES	80.02	23.22	155.20	0.00	0.00	228.44	6.67	42.41	4.07	0.01	507.8
s4	CryptoMiniSat (native XOR)	76.38	22.97	149.10	1.66	10.85	127.17	14.01	44.32	4.47	0.01	436.0
s4	Z3 (And/Xor)	78.62	21.30	149.56	24.86	1305.62	3132.67	18.93	40.90	4.16	0.01	4747.2
s4	F4-style prototype	84.55	23.38	149.96	0.00	0.00	4936.27	0.00	0.00	0.00	0.00	5175.4
s4	F5B prototype	76.36	21.08	145.25	4.74	6.59	5245.77	0.00	0.00	0.00	0.00	5470.5

- Setup failures are inputs with no planted triple (2 of 4 cells at s = 2 and 3, 1 of 4 at s = 4), as in m83; they are not attempts, and their setup cost is not in the per-attempt averages.
- Timeouts are charged their full deadline (about 5 s of solver time) to the engine. F4-style and F5B time out at s >= 3 and Z3 reaches its deadline at s = 4 on some cells, so their totals at larger s are set by the deadline, not by an algorithm's completion time.
- **Deviation from m83**: this host's solver stage is faster than m83's (section 8), and several cells near the 5 s deadline completed here that m83 recorded as timeouts: Z3 at s = 4 and one F5B cell at s = 3 (9 cells over 3 repetitions, listed in `results/verify_report.json`). Cells whose status changed between repetitions are named in the table. A timeout at a fixed 5 s deadline is a property of the host speed as much as of the engine.
- At s = 4 the solver stage is no longer small next to equation build, and CryptoMiniSat's total is below FES's (436.0 ms against 507.8 ms per attempt), as in m83's solver-stage ranking. No natural relation was found at any size.

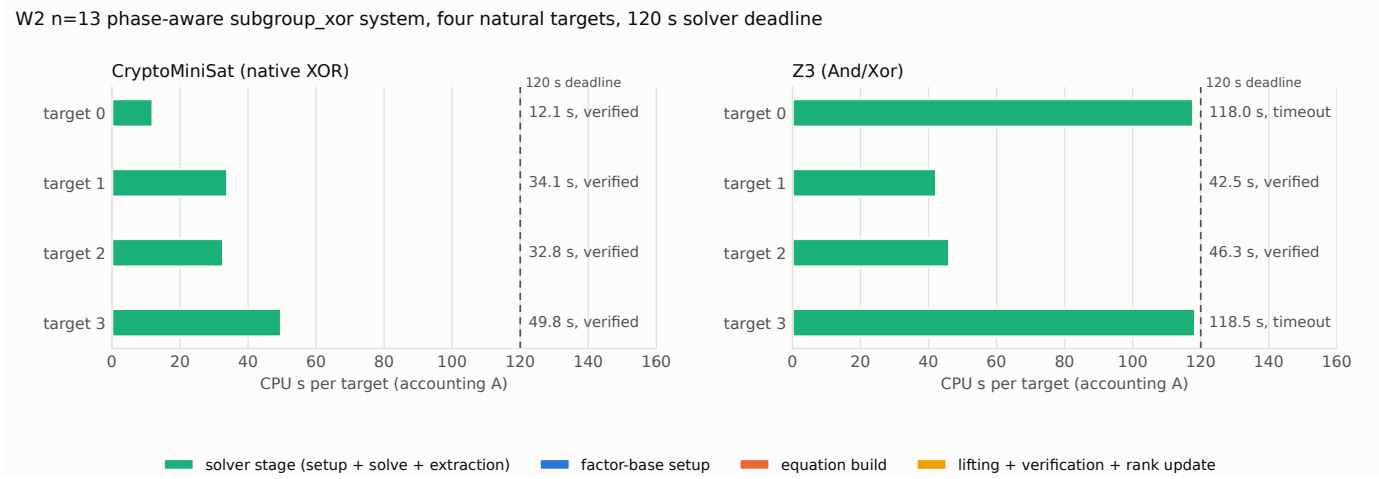
7. W2: n = 13 phase-aware systems, CryptoMiniSat (native XOR) against Z3 (And/Xor)

The encoder of `research/frobenius_quotient_sat_20260930` is unchanged (formulation `subgroup_xor`, phases unknown, s = 4, seed 20260930; 3627 Boolean variables, 7927 clauses, 1041 XOR constraints, 2535 AND gates). The four targets are the four *random* (natural) targets that harness draws; their coordinates and the factor-space basis were checked equal to the recorded `results/n13_phase_b300.jsonl` meta before the

run. Z3 receives exactly the clauses and XOR constraints the encoder produced (clause to `Or`, XOR to an `Xor` tree). The 120 s aggregate deadline starts when the solver stage starts, so it includes Z3's translation.

Engine	target	outcome	T_core CPU s	solver stage CPU s	f_s	solve wall s	rejected tuples	new independent natural relation	peak RSS MB
CryptoMiniSat (native XOR)	0	verified	12.14	12.11	99.7%	12.28	0	1	56
CryptoMiniSat (native XOR)	1	verified	34.14	34.10	99.9%	34.51	0	1	113
CryptoMiniSat (native XOR)	2	verified	32.76	32.72	99.9%	33.17	0	1	109
CryptoMiniSat (native XOR)	3	verified	49.80	49.76	99.9%	50.35	0	1	95
Z3 (And/Xor)	0	timeout	118.01	117.98	100.0%	118.91	None	0	121
Z3 (And/Xor)	1	verified	42.47	42.43	99.9%	42.24	0	1	101
Z3 (And/Xor)	2	verified	46.34	46.31	99.9%	45.72	0	1	115
Z3 (And/Xor)	3	timeout	118.54	118.52	100.0%	118.89	None	0	131

Engine (W2, 4 natural targets)	verified	timeouts	unsat	new independent relations	T_core CPU s (sum)	f_s	T_core CPU s per new independent relation	T_core CPU s per attempt
CryptoMiniSat (native XOR)	4/4	0	0	4 (rank 4)	128.8	99.89%	32.2	32.2
Z3 (And/Xor)	2/4	2	0	2 (rank 2)	325.4	99.97%	162.7	81.3



- **The dominant component flips with the workload.** At n = 13 the solver stage is 99.9% (CryptoMiniSat) and 100.0% (Z3) of the charged cost, against 0.5%-66% for the six-variable m = 83 systems of W1a. Everything outside the solver together costs 139.7 ms (CryptoMiniSat) and 112.8 ms (Z3) of CPU per four targets, so here engine choice is essentially the whole cost. This compares workload scale, not engine merit.
- **Here the metric is finite.** These targets are natural and decompose, so CryptoMiniSat produced 4 independent verified natural relations (rank 4 over Z/2003, rows re-reduced independently by `verify.py`) at 32.2 CPU s per relation. Z3 produced 2 in 325.4 CPU s: 162.7 CPU s per relation, i.e. 5.1x CryptoMiniSat's. That figure is a **lower bound**: the two timed-out targets are charged the 118-120 s they consumed, and their true cost is higher (right-censored).
- On the two targets both engines solved, Z3 took 1.2x and 1.4x CryptoMiniSat's time; on the other two CryptoMiniSat took 12 s and 50 s and Z3 reached its deadline. Four targets, one trial each, and an unconfigurable CryptoMiniSat seed: the per-target spread is as large as the engine gap on the solved pair, so only the coarse statement is supported (CryptoMiniSat with native XOR solved all four, Z3 with Xor terms two).
- The comparison pits native XOR against XOR flattened to terms; it is not a statement about SAT solvers in general. The Z3 runs saw 1-minute load averages of about 2-3 against 1.3-2.0 during CryptoMiniSat's; CPU seconds are reported, but cache and frequency effects of a shared host are not controlled.
- This does not repeat Lead 1's paired phase-on/phase-off comparison on the same harness; it varies the engine, not the formulation.

8. Bridge to the m83 numbers, and deviations

Engine	m83 published median solver-stage wall s (recomputed from archive)	this host, same definition (incl. library import), wall s	this host, library load excluded, wall s	this host / m83
FES	0.000335	0.000236	0.000236	0.70x
CryptoMiniSat (native XOR)	0.028128	0.011433	0.009825	0.41x
Z3 (And/Xor)	0.212127	0.103238	0.078695	0.49x
F4-style prototype	0.171046	0.136095	0.136095	0.80x
F5B prototype	0.149750	0.089875	0.086955	0.60x

- The m83 solver-stage timer wrapped the library import (`import z3`, `import pycryptosat`, the F5B module import) inside the timed region, so its published medians include library load. Here `engine_load` is a separate stage; the third column restores the m83 definition and the fourth removes the load.

- This host's solver stages are 0.4x-0.8x of the m83 run's. The m83 receipts record Python 3.12.14 (a Clang build), Z3 4.15.3 and `Linux-6.18.44` but no CPU model, so whether the hardware differed is not established; the interpreter and Z3 builds certainly differ. This run: Python 3.13.16 (GCC build), Z3 5.1.0, `pycryptosat` 5.16.0, an Intel Xeon at 2.10 GHz under a Linux kernel with the same 6.18.44 base. Absolute seconds are not comparable across runs; the ratios between stages within one run are the claim.
- Repetition spread (T_core sums over the 12 W1a inputs; the host was shared):

Engine (W1a, 12 inputs summed)	T_core CPU s, min over reps	median	max	T_solver CPU s, min	max	max 1-min load average at a child start
FES	0.852	0.951	0.961	0.0039	0.0046	2.32
CryptoMiniSat (native XOR)	0.958	0.986	1.084	0.0979	0.1070	2.32
Z3 (And/Xor)	1.718	1.858	1.858	0.8672	0.9562	2.32
F4-style prototype	2.487	2.537	2.549	1.6337	1.6686	2.32
F5B prototype	1.887	2.010	2.130	1.0386	1.2093	2.32

- `PYTHONPYCACHEPREFIX` pointed at a warmed cache outside the repository, so bytecode compilation is not in any stage; a warm-up child ran before the first measurement.

9. Preregistered predictions, checked

1. *Natural verified relations are 0 in every W1 cell.* Held: 0 of 8 (W1a) and 0 of the completed natural attempts of W1b for every engine.
2. *`fb_setup` exceeds `eq_build` and, for FES and CryptoMiniSat, `T_solver`.* Held: 53.1 ms against 7.4 ms, and against solver stages of 0.38 and 8.31 ms.
3. *`f_s` below 5% for FES, below 50% for CryptoMiniSat, above 50% for Z3 (accounting A).* Held, narrowly for Z3: 0.5%, 9.9%, 50.5% (wall: 50.4%).
4. *Orderings by `T_core` and by `T_solver` agree, with the ratio shrinking sharply.* Held under A (tau 1.000), B and the post hoc steady state; **not held under C**, where Z3 and F5B swap (tau 0.800). The shrinkage is accounting-dependent: max/min over engines is 365.4x (solver stage), 2.7x (A), 5.2x (B), 2.3x (C) and 19.9x (post hoc).
5. *In W2 the solver fraction exceeds 95% for both engines.* Held: 99.9% and 100.0%.

Amdahl thresholds declared in the protocol (`g < 1.25` negligible, `1.25 <= g < 2` minor, `g >= 2` majority): FES 1.00x and CryptoMiniSat 1.11x are negligible; Z3 2.02x, F5B 2.22x and F4-style 2.91x are majority-solver (accounting A).

10. Scope limits

- Stage diagnostic only. No discrete logarithm, no rank-complete matrix, no matched rho, no `S`, no ratio to the floor, no ECC2K-130 or m=131 transfer, and no ECDLP speedup claim.
- The two m=83 factor bases have 2 and 3 Frobenius-orbit columns; the rank cap is 2 or 3. The fixed phase triple, six payload variables and 4-6 usable lifts make the expected natural yield about 1e-22 per attempt, so this workload cannot show a relation yield for any engine.
- All engines other than CryptoMiniSat and Z3 are the pinned **Python prototypes** of the m83 matrix, not Magma F4/F5 or msolve (unavailable here). The factor-base, equation-build, lifting and verification stages are pure-Python field and curve arithmetic. The stage fractions are properties of that implementation: a native pipeline shrinks the non-solver stages and raises every solver share. Under `AGENTS.md` ("no Python") these results must be re-measured natively before they inform a performance claim; the frozen workload and engines are legacy Python by assignment.
- CryptoMiniSat's binding exposes no seed; Z3 uses seed 13. Timings are single-host, shared-CPU, three repetitions for W1 and a single trial per cell for W2.
- Single-thread CPU seconds (`process_time`) are primary, wall seconds are reported alongside; both agree closely here (0.962 s wall against 0.951 s CPU for FES, summed over the inputs).
- No canonical scoreboard, leaderboard or lab-browser panel changes: the work lands no whole-pipeline measurement, reference, curve or verdict change (`AGENTS.md` sections 7-7c).

11. Reproduction

```
cd research/notes/ecc2k130/solver_total_cost_20261011
export PYTHONHASHSEED=0 PYTHONPYCACHEPREFIX=/some/scratch/pycache
python3 preflight.py
python3 -m unittest test_rank && python3 test_replication.py
timeout 1800 python3 run_w1.py --out results/w1 --reps 3
timeout 1700 python3 run_w2.py --out results/w2
python3 summarize.py && python3 make_figures.py && python3 build_report.py && python3 verify.py
```

Outputs: `results/w1/receipts.jsonl`, `results/w2/receipts.jsonl`, `results/summary.json`, `results/tables.md`, `results/verify_report.json`, `figures/*.svg|png`, `report.pdf`.